Named KiCad buses show complete grouped routes; member pin stubs carry exact net labels. RESET_N is an explicit Pico / Z80 / GAL three-way wire junction.

REGULATED 5V INPUT

1N5819

PWR_FLAG

MCP TEST

RESET_N

MEMORY BOARD

AS6C1008-55PCN

ADDRESS

DATA

PICO_TO_GAL

Z84C2520FEC

HCT_TO_CPU

CPU_TO_GAL

GAL_TO_HCT

HCT_TO_SRAM

SN74HCT541N

CORE BOARD

PICO_DATA

PICO_TO_HCT

HCT_TO_MCP

CPU_MONITOR

PICO_MONITOR

ADDRESS_CONTROL

DATA_CONTROL

MCP_GPA

MCP_GPB

SN74HCT245N

SN74HCT245N HIGH

SN74HCT245N LOW

SN74HCT244N

MCP23S17-E/SP

PERIPHERAL BOARD

Raspberry Pi Pico 2 W

SN74LVC1G08DBVR

Startup guide: R1–R16 = 5V pull-ups, R17–R23 = 3V3 pull-ups, R24–R29 = 3V3-side pull-downs.

R1 10k BUSREQ_N
R2 10k BUSACK_N
R3 10k MREQ_N
R4 10k IOREQ_N
R5 10k RD_N
R6 10k WR_N
R7 10k MCP_SO
R8 10k SRAM_CE_N
R9 10k SRAM_OE_N
R10 10k SRAM_WE_N
R11 10k WAIT_N
R12 10k INT_N
R13 10k NMI_N
R14 10k SRAM_WE_PRE_N
R15 10k SRAM_OE_PRE_N
R16 10k SRAM_CE_PRE_N
R17 10k PICO_BUSREQ_N
R18 10k PICO_CE_N
R19 10k DATA_OE_N
R20 10k ADDR_OE_N
R21 10k PICO_SPI_CS_N
R22 10k PICO_WE_N
R23 10k PICO_OE_N
R24 10k PICO_CLK
R25 10k RESET_N
R26 10k DATA_DIR
R27 10k ADDR_DIR
R28 10k PICO_SPI_SCK
R29 10k PICO_SPI_MOSI

C1 100n Z80
C2 100n CPU SRAM
C3 100n GAL
C4 100n HCT541
C5 100n HCT245
C6 100n CPU HCT245
C7 100n CPU LVC244
C8 100n CPU MCP
C9 100n CPU VCCA
C10 100n CPU VCCB
C11 10u Memory Board
C12 10u Core Board
C13 10u Peripheral Board
C14 47u Supply Entry

Native KiCad 10 schematic generated from the verified README pin map

Z80ROMlessSBC

Sheet: /

File: z80_romless_sbc.kicad_sch

Title: Z80ROMlessSBC — Exact Breadboard Electrical Schematic

Size: A0	Date: 2026-05-13	Rev: 1
KiCad E.D.A. 10.0.5		Id: 1/1